

Somatic Modalities as Complementary Approaches to Chronic Pain Management

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Chronic pain is a historically misunderstood, difficult to treat patient experience complicated by a historical dualistic diagnostic model that understood ailments as being either rooted in the body or the mind did not account for environmental, sociological, or other external factors. (Gatchel, Peters, Fuchs, & Turk, 2007) A multimodal understanding has since been adopted starting in the 1980s – considered a biopsychosocial model and a new medical subspecialty called psychoneuroimmunology (Patt, 2004). It can be difficult to untangle if chronic pain causes psychological distress or if psychological distress causes existing pain to become chronic, but there is evidence that chronic pain typically has both physical and psychological factors. Chronic pain affects a substantial portion of the population, with estimates as high as one in fourteen adults experiencing “high impact” chronic pain. (Wang, Aaron, Attal, & Colloca, 2025)

A common view in somatic psychology is that the body is reacting to something in the environment or mind that the patient is often not presently aware of. That can be a past memory or something currently in the environment that is causing the patient pain or distress. Part of the practice is to bring awareness to that communication from the body to bring about resolution in some way. This can be anxiety in anticipation of future pain that causes muscle tension, which causes muscle strain, which causes its own separate pain in addition to a medical condition which is the source of a separate pain. By bringing awareness to the body, working through that physical reaction and the anxiety itself, the additional pain can be lessened or eliminated completely, leaving only the medical pain behind (which can perhaps be lessened since the body and mind are now relaxed). Studies have linked a reduction in catastrophizing to a reduction in chronic pain, which can be accomplished through practices like meditation, Tai Chi, and CBT. (Stroud, et al, 2019)

While the mechanism of mind-body may not be fully understood, the efficacy has been recognized as early as 1991 when the National Institutes of Health established the Office of Alternative medicine, later renamed the National center of Complementary and Alternative Medicine in 1998 and renamed again in 2014 to the National Center for Complementary and Integrative Health. (Stroud, et al, 2019) Additionally, the Veterans Administration has implemented a multimodal integrative approach and Medicaid plans have begun covering acupuncture and psychological interventions for back pain. (Stroud, et al, 2019)

When somatic practices can't treat the cause of pain, they can help with the body's ability to cope with it. Relaxation practices like meditation, Yoga, Tai Chi, and Qigong have been shown to lower the body's production of adrenaline, cortisol, neuropeptides, which can decrease overall health, increase stress responses and tension, and impact the

immune system. They also increase the production of endorphins, which help decrease pain sensitivity. (Patt, 2004) A study conducted by neurochemists at the Maharishi International University found that after a four-month long meditation practice, cortisol levels in participants were reduced by 15 percent. Another study found a 25% reduction. (Patt, 2004)

Somatic modalities are many and it would be surprising if there wasn't one suitable for a patient. Commonly studied and available modalities are meditation, relaxation techniques (including progressive muscle relaxation, controlled breathing, autogenic training, and visualization), Yoga, Qigong, and Tai Chi. Additionally, meditation, relaxation, and yoga, are more open to self-study., allowing patients a certain level of independence, control, and self-determination in a circumstance where many feel helpless. While Qi Gong and Tai Chi are typically practiced under the guidance of a professional, they aren't what are typically considered a behavior or mental health engagement which can feel more inviting. With such a diversity of modalities and accessibility, mind-body practices can be much more accessible for patients.

Mindfulness-based approaches are focused on developing an awareness of the present moment and what is happening in it with a non-judgmental experience. In a meta-analysis of randomly controlled trials, reduction in pain intensity was observed, though with a small effect, though some studies have found that compared as a stand-alone treatment with CBT or standard care alone, mindfulness-based stress reduction had more significant results for patients with chronic lower back pain than CBT and standard care. (Wang, Aaron, Attal, & Colloca, 2025) They also showed improvements in functional imitations for the following 6 months, along with patients from the CBT group. ((Wang, Aaron, Attal, & Colloca, 2025)

Movement based approaches include Yoga, Tai Chi, and Qigong. While Tai Chi, Yoga, and Qigong share some similarities – they each practice a selection of body movements or postures, breathing exercises, and meditations, it's important to recognize a cultural, philosophical and spiritual difference between practices. Qigong is an ancient Chinese practice that uses its techniques to balance “qi” or “chi,” which is the body's life energy. Tai chi is a traditional Chinese practice that uses flowing movements for balance and mental focus, the practice itself a moving meditation. Yoga comes from ancient India and has many styles and branches, though the most popular in the west is Hatha Yoga.

In a meta-analysis of 16 randomly controlled trials, a 12-week yoga practice improved patients' chronic pain significantly and produced neuroanatomical changes including neuroprotective affects and alterations in gray matter volume. (Wang, Aaron, Attal, & Colloca, 2025) Though the physiological effects are not yet understood. Similarly

Qigong showed significant reductions in chronic pain, physical function, and life satisfaction, in comparison to placebo controls as well as “sham Qigong” (Qigong practiced without breathing or meditation aspects) Patients with osteoarthritis and chronic low back pain practicing Tai Chi also showed significant reductions in chronic pain levels while patients with fibromyalgia found relief from chronic pain, better sleep, and reduced fatigue. (Wang, Aaron, Attal, & Colloca, 2025)

More extensive studies are necessary because most studies tend to focus on short term results and long-term studies seem to be unavailable. Additionally, the functional mechanism of efficacy is not yet understood behind these different methodologies, although the same is often true for pharmacological solutions. (Topal, 2019) It may be that somatic practices, especially those with social and spiritual aspects, may have their benefits rooted in the placebo effect (whose efficacy is equal to or surpasses medication) or in those social and spiritual aspects. (Stroud, et al, 2019)

However, the best success for patients has been seen in programs with a multimodal approach rather than programs that take any single approach, whether that is a pharmacological approach, a self-management approach, or a purely somatic approach. A study conducted by Kurt Kroenke in 2018 found that, when comparing analgesics, mood treatments, and self-management to self-management only, one-third of participants receiving the usual care worsened over 12 months compared to one-sixth of participants in the active arm. (Stroud, et al, 2019)

While there are more studies and research to do, it’s clear that somatic therapy has a place in the treatment of chronic pain management in concert with more traditional medical methods. Especially considering open-label placebos have a nearly equivalent efficacy rate as pharmacological methods (Wang, Aaron, Attal, & Colloca, 2025) and over 75% of patients receiving the 10 most prescribed prescriptions do not experience the intended effects. (Topal, 2019)

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